

Reset the Entryway SOP (1x Episode: coats, shoes, keys, bag, mat, in situ)

One episode resets the whole entryway at the bench where these things live. The two coats go first. Then the two shoes. Then the keys. Then the bag. The mat is squared last. The episode ends when the bench top is clear, every thing hangs or sits where it belongs, and the mat runs straight in front of the bench. Never split the coats, the shoes, the keys, the bag, and the mat into two episodes.

The base is **passive**: it has no drive of its own, so it is pushed by hand up to the bench in front of the hook board and locked there, and nothing is carried away to another room. Everything the episode touches is already there when recording starts: the bench, the hook board above it with its three empty hooks, the two coats, the two loose shoes, the keys, the bag, the empty bowl at the back right, the empty shoe rack on the floor at the left end, and the mat on the floor in front of the bench.

The entryway is worked **as found**. The three hooks are empty. The bowl is empty. The shoe rack is empty. The bench top is clear but for the two coats, the two shoes, the keys, and the bag. The mat lies on the floor crooked. The episode ends with two coats on the two coat hooks, two shoes paired on the rack, the keys in the bowl, the bag on the bag hook, the bench top clear, and the mat square.

This is an in-situ task, and four things follow from that. First, **everything happens at this bench**: nothing is carried away and nothing new is brought in. Second, **every carry goes along the front lane, low and level**: a thing is lifted just off what it lies on, brought forward into the lane, and carried along the lane close over the bench. The only lifts that go up are the coat loops and the bag strap going onto their hooks, and the only reaches that go down are the shoes going onto the rack and the mat. Third, **the fixtures never move**: nothing leans on, pushes, drags, or lifts the bench, the hook board, a hook, the bowl, or the shoe rack. Fourth, **nothing is dragged**: every pick lifts the thing straight up off the surface before it moves anywhere. The mat is the one thing that is never lifted. It is pushed flat along the floor.

Things go on the bench, on a hook, in the bowl, or on the rack, and nowhere else. Nothing is ever left on the floor but the shoes on the rack and the mat. A thing that lands on the floor is a drop. It is not picked back up. It is reported.

The bench is set up in one of three ways. The hook board, the bowl, the shoe rack, the mat, the shoes, the keys, and the bag keep their places. Only the coat stack moves, and the shoes make room for it in Config M.

- **Config L**: the coat stack lies at the **front left**. The shoes lie loose at the front middle.
- **Config M**: the coat stack lies at the **front middle**. The shoes lie loose at the front left.

- **Config R:** the coat stack lies at the **front right, at the right end of the bench past the bag**. The shoes lie loose at the front middle.

One config per episode, chosen before recording and never changed mid-episode. Where a step depends on the setup it says so on an **IF** line. Look at the bench and follow the line that matches.

Same-side rule: the gripper on the coat stack's side is the one that lifts each coat, and the **left gripper** is the one that hangs it, because both coat hooks are on its side. In Config M the stack lies in the shared middle and the left gripper lifts. **Hand-over rule:** in Config R the coat hooks are too far for the right gripper, so the right gripper lifts the coat and hands it over to the left gripper above the middle of the front lane. Nothing else is ever passed from one gripper to the other.

The other roles are fixed: the **left gripper** takes both shoes to the rack, because the rack is at its end of the bench. The **right gripper** takes the keys to the bowl and hangs the bag, because the bowl and the bag hook are on its side.

The order never changes: **hang the first coat, hang the second coat, put the first shoe on the rack, pair the second shoe beside it, put the keys in the bowl, hang the bag, square the mat**. Nothing starts until the thing before it is done and let go.

Setup

Complete the base positioning and all the checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Base positioning

The base is **passive**. It has no drive of its own, it is pushed into place by hand, and it is never steered, nudged, or repositioned once recording starts. It is parked once, before recording, and does not move again until the episode is over.

1. Push the base by hand up to the bench, behind the mat, and stop it **square to the front edge of the bench**, so the front edge runs straight across the frame and neither end sits nearer than the other.
2. Stop it **centered on the bench**, so the left end and the right end of the bench are the same distance from the middle of the base.
3. Stop it **close enough** that the **left gripper** reaches coat hook one and the **right gripper** reaches the bowl at the back right without either arm extending, and **far enough** that no

arm, wrist, or part of the base touches the bench front, the hook board, or the mat while both arms work.

4. Check the **height**: with the base parked, the **left gripper** lifts a coat off the bench by its loop until its hem just clears the bench, brings it forward into the front lane, and carries it to coat hook one, without the wrist or the forearm touching the hook board.
5. Check the **hooks**: the **left gripper** puts a loop over each coat hook and lets go, and the **right gripper** puts the bag strap over the bag hook and lets go, without extending and without touching the hook board or the other arm.
6. Check the **low reaches**: the **left gripper** reaches the shelf of the shoe rack, and each gripper reaches the corner of the mat on its own side, without extending and without the base touching either one.
7. Check the **bench**: the gripper on the coat stack's side reaches the stack, the **left gripper** reaches both shoes, and the **right gripper** reaches the keys, the bag, and the bowl, all without extending and without knocking anything over.
8. Lock or brake the base. Push it firmly once by hand: it must not roll, creep, or turn.
9. If any of lines 1 to 7 fails, push the base to a new park by hand and start again at line 1. Do not work a bench the arms cannot reach comfortably.

The base stays locked and still for the whole episode. Nothing moves it: no arm leans on the bench or the hook board hard enough to shift it, nothing and nobody touches it, and it is never repositioned mid-task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the middle of the bench and its frame includes the whole bench top, the hook board with all three hooks, the two coats, the two shoes, the keys, the bag, the bowl, the shoe rack, and the whole mat.
3. The camera reads the bench from the front, so where a thing lies and whether it lies flat and square are readable.
4. The camera reads each hook from the front, so whether a loop or a strap sits all the way over it is readable.
5. The camera reads into the bowl from the front, so whether the keys are in it is readable.
6. The camera reads the shelf of the shoe rack, so how the two shoes stand and whether they are paired is readable.
7. The camera reads the front edge of the bench and the near edge of the mat in the same frame, so whether the mat runs straight is readable.
8. Both arms are at home with grippers open.
9. The **left arm** reaches both coat hooks, the shoe spot, the rack shelf, the left corner of the mat, and the middle of the front lane, without extending to a joint limit.

10. The **right arm** reaches the key spot, the bag spot, the bowl, the bag hook, the right corner of the mat, and the middle of the front lane, without extending to a joint limit.
11. The gripper on the coat stack's side reaches the stack where this config puts it without extending.
12. A gripper lifts a thing just off what it lies on and carries it low and level along the front lane. It never carries one across the back of the bench and never over the bowl.
13. The two arms do not collide. An arm working at a hook stays out of the other arm's path along the front lane.
14. If a place cannot be reached, re-park the base by the Base positioning steps until lines 9 to 13 hold.

Materials checklist

1. The **bench** is one long flat top set against the wall under the hook board. Nothing moves it, leans on it, or stands anything on its edge.
2. The bench top is dry, clean, and clear but for the two coats, the two shoes, the keys, and the bag.
3. The **hook board** is fixed to the wall above the back of the bench. It does not move, turn, or come loose. Nothing pushes it or hangs on it.
4. The hook board carries **three hooks** in a row: **coat hook one** at the left, **coat hook two** next to it, and the **bag hook** at the right end. All three are empty at the start.
5. The hooks stand out far enough that a loop or a strap sits all the way over one and stays there.
6. Each **coat** is a jacket with a hanging **loop** sewn inside the collar. The loop stands up when the coat lies on the bench.
7. The two coats lie flat at the **coat spot** this episode's config puts them, one on top of the other, each with its loop up and clear: the **front left in Config L**, the **front middle in Config M**, the **right end of the bench past the bag in Config R**.
8. Neither coat has anything in its pockets, and no belt, cord, or sleeve hangs off the bench edge.
9. The **shoes** are one pair, a left shoe and a right shoe of the same kind. They lie loose at the **shoe spot**, apart from each other and not paired: the **front middle in Config L and R**, the **front left in Config M**.
10. Both shoes are dry and clean underneath, so they leave no mark on the bench or the rack.
11. The **shoe rack** is a low open rack standing on the floor at the left end of the bench, clear of it. It has one shelf. It is empty at the start. Nothing moves it, tips it, or stands on it.
12. The rack shelf is wide enough for the two shoes to stand side by side with a finger's width between them.

13. The **keys** are one set on one ring. They lie at the **key spot**, the front right of the bench. No key is loose off the ring.
14. The **bowl** is a shallow bowl standing at the back right of the bench top. It is empty at the start. It is wide enough and deep enough that the whole key set lies inside it. Nothing moves it, lifts it, or slides it.
15. The **bag** is one bag with one **strap**. It lies at the **bag spot**, the front right of the bench past the keys, with its strap laid out clear and not under the bag.
16. The bag is closed and holds nothing loose that can fall out when it hangs.
17. The **mat** is one flat mat lying on the floor between the base and the bench. It starts crooked, turned off square to the bench front edge, but fully flat with no corner folded under.
18. The mat slides on the floor when it is pushed at its edge, and it does not bunch up or roll.
19. The floor around the bench is dry and clear, so nothing is in the mat's way.

Workspace layout

Nothing is marked or taped out. Every place below is judged by eye from the bench, the hook board, and the mat themselves.

- **Above the bench, left to right:**
 - **Coat hook one**, at the left of the hook board. Empty at the start, holds the first coat at the end. Never moved.
 - **Coat hook two**, next to coat hook one. Empty at the start, holds the second coat at the end. Never moved.
 - **Bag hook**, at the right end of the hook board. Empty at the start, holds the bag at the end. Never moved.
- **Back row of the bench top:**
 - **Bowl spot**, the back right. The bowl stands here, empty at the start and holding the keys at the end.
- **Front row of the bench top:**
 - **Coat spot**: where the two coats lie in one flat stack, front left in Config L, front middle in Config M, right end past the bag in Config R. Empty by the end of Step 2.
 - **Shoe spot**: where the two loose shoes lie, front middle in Config L and R, front left in Config M. Empty by the end of Step 4.
 - **Key spot**, front right. The keys lie here. Empty by the end of Step 5.
 - **Bag spot**, front right past the keys. The bag lies here. Empty by the end of Step 6.
- **On the floor:**
 - **Shoe rack**, at the left end of the bench, clear of it. Empty at the start, holds the paired shoes at the end.

- **Mat**, between the base and the bench. Starts crooked and ends square.
- **Front lane**: the clear strip of bench top along the front edge, from one end of the bench to the other. Every carry goes along it. Nothing is ever put down in it.

No carry passes across the back of the bench or over the bowl, and nothing travels to the far side of the bench.

Arm assignments

- **The gripper on the coat stack's side lifts each coat**: the left gripper in Config L and M, the right gripper in Config R. **The left gripper hangs every coat**, on coat hook one and then on coat hook two. In Config R the right gripper hands the coat over to the left gripper above the middle of the front lane.
- **The left gripper owns the shoes**. It lifts each shoe from the shoe spot, carries it along the lane to the left end, and stands it on the rack shelf.
- **The right gripper owns the keys and the bag**. It lifts the keys by the ring and lowers them into the bowl, then lifts the bag by its strap and hangs it on the bag hook.
- **The mat is pushed by the gripper on the corner's side**: the left gripper for the left corner, the right gripper for the right corner, and the right gripper sets the gap.
- Only one thing is held at a time. No grip is changed while a thing is off the bench, except the Config R coat hand-over. Every thing is put down or let go under control.

Vocabulary

- **In situ**: the entryway is worked where it is. Nothing is carried away to another room, and nothing about the entryway is rearranged for the task.
- **Passive base**: the base has no drive. It is pushed by hand to its park before recording, locked there, and it does not move at all during the episode.
- **Config**: which of the three coat spots this episode uses. It is set before recording and never changes mid-episode.
- **Coat**: one of the two jackets at the coat spot. Both are hung before anything else moves.
- **Loop**: the small sewn loop inside a coat's collar. It is the only part of a coat a gripper ever closes on.
- **Strap**: the long band on the bag. It is the only part of the bag a gripper ever closes on.
- **Top of the stack**: the coat lying highest with nothing on it. The top one is always taken. One is never pulled out from under the other.
- **Just off**: lifted only until the thing's lowest part clears what it was lying on, and no higher. For a coat or the bag hanging from a gripper that means the hem or the bottom of the bag just clears the bench.

- **Low carry:** the thing is lifted just off, brought forward into the front lane, and carried along the lane close over the bench and level. It never goes across the back of the bench and never over the bowl.
- **Front lane:** the clear strip of bench top along the front edge. Every carry goes along it. Nothing is put down in it.
- **Hand over:** in Config R only. The right gripper holds the coat still by its loop above the middle of the front lane, the left gripper closes on the loop beside the right gripper's jaws, and then the right gripper opens. The coat is never let go between the two grips.
- **Hanging right:** the loop or the strap sits all the way over the hook, the thing hangs straight down, and it does not swing, slide off, or lift off the hook when the gripper lets go.
- **Hanging clear:** the hung coat or bag hangs free of the bench top, free of the wall behind it, and free of the thing hanging next to it. No sleeve, hem, or strap rests on the bench.
- **Standing right on the rack:** the shoe stands flat on the rack shelf, upright on its own sole, fully on the shelf with no part hanging over the front edge, and it does not fall or rock when the gripper lets go.
- **Paired:** the two shoes stand side by side on the rack shelf, both facing the same way, their heels level with each other, and a finger's width between them. Not one in front of the other. Not one on top of the other.
- **In the bowl:** the whole key set lies inside the bowl, below its rim, with no key hanging over the edge and no key outside the bowl.
- **Square:** the near edge of the mat runs straight across, parallel to the front edge of the bench, and both ends of that edge are the same distance from the bench.
- **Push flat:** the closed gripper touches the mat's edge close to the floor and pushes it along the floor. It never lifts the mat, never digs under it, and never bunches it up. **Unvalidated:** whether a gripper reaches the mat's edge at floor level from the parked base is to be checked at the station.
- **Drop:** a coat, a shoe, the keys, or the bag that falls on the floor. It is not picked back up. It is reported.
- **Loose thing:** a thing that ends up on the bench outside its spot, or half on a hook, or beside the bowl. It is left where it lands and cleared in the reset, never on camera.
- **Back and clear:** the arm is drawn back so that no part of it is over the bench top, the front lane, a hook, or the mat.

Handling standard

- Move exactly one thing at a time. Nothing is passed from one gripper to the other, except the coat hand-over in Config R.
- Every lift comes straight up, just off the surface, before the thing moves anywhere.
- Every carry is a low carry along the front lane.

- Nothing is put down anywhere but its own place in the step being done, and never in the front lane.
- Nothing rests on a fixture. No gripper, wrist, or forearm leans on the bench, the hook board, a hook, the bowl, or the shoe rack, and no push is ever hard enough to move any of them.

Steps

Steps 1 and 2 depend on the config: the gripper on the coat stack's side lifts the coat, and in Config R it hands the coat over to the left gripper. Every other step is the same in all three.

Step 1: Hang the first coat

Goal: the first coat hangs on coat hook one, and the coat spot holds one coat.

1.1 Lift the top coat by its loop

- **IF Config L or M:** the **left gripper** closes on the loop of the top coat at the coat spot, lifts the coat straight up off the stack until its hem just clears the coat under it, and brings it forward into the front lane.
- **IF Config R:** the **right gripper** closes on the loop of the top coat at the coat spot, lifts the coat straight up off the stack until its hem just clears the coat under it, brings it forward into the front lane, and carries it in a low carry to the middle of the lane. There it **hands over:** the **right gripper** holds still, the **left gripper** closes on the loop beside it, and the **right gripper** opens and comes back and clear.

Then, in all three:

- Do not swing the coat. Do not drag it along the bench.

Check: the coat hangs straight down from its loop in the **left gripper**, and the coat under it has not come with it and has not slid. If both coats came up together, the gripper that lifted lays both back on the coat spot, squares them, and takes the top one again by its loop.

1.2 Carry it to the hook

- The **left gripper** carries the coat along the front lane to coat hook one in a low carry, keeping it hanging straight the whole way.
- Do not carry it across the back of the bench and do not carry it over the bowl.
- Bring it in front of the hook, then lift the loop straight up, level with the hook.

Check: the coat hangs straight and still under the **left gripper**, in front of coat hook one. No sleeve is caught on the bench, on the bag, or on the other arm. If the coat swings, hold it still and let it settle before going on.

1.3 Put the loop on the hook

- The **left gripper** moves the loop straight back over the hook, so the hook goes through it.

- Lower the loop until it sits all the way down on the hook and the coat takes its own weight.
- Open the gripper and draw it back away from the coat.

Check: the coat is hanging right and hanging clear. The loop sits all the way over the hook, the coat hangs straight down, it does not swing or lift off, and no part of it rests on the bench. If the loop sits on the tip of the hook or the coat rests on the bench, the **left gripper** closes on the loop again, lifts it just off the hook, and puts it on again.

Expected state: one coat hangs on coat hook one. One coat lies at the coat spot. The other two hooks are empty. Both grippers are back and clear.

Step 2: Hang the second coat

Goal: the second coat hangs on coat hook two, and the coat spot is empty.

2.1 Lift the second coat by its loop

- **IF Config L or M:** the **left gripper** closes on the loop of the coat left at the coat spot, lifts it straight up until its hem just clears the bench, and brings it forward into the front lane.
- **IF Config R:** the **right gripper** closes on the loop of the coat left at the coat spot, lifts it straight up until its hem just clears the bench, brings it forward into the front lane, carries it in a low carry to the middle of the lane, and **hands over** to the **left gripper** the same way as in Step 1.1.

Check: the coat hangs straight down from its loop in the **left gripper** and the coat spot is empty. Nothing came up with it. If the coat hangs by a sleeve or a hem instead of the loop, the gripper that lifted lays it back down flat, closes on the loop, and lifts it again.

2.2 Carry it to the hook

- The **left gripper** carries the coat along the front lane to coat hook two in a low carry, keeping it hanging straight the whole way.
- Do not carry it across the back of the bench and do not carry it over the bowl.
- Bring it in front of the hook, then lift the loop straight up, level with the hook.

Check: the coat hangs straight and still in front of coat hook two, clear of the first coat. If it brushes the first coat, the **left gripper** moves it a little away and holds it still before going on.

2.3 Put the loop on the hook

- The **left gripper** moves the loop straight back over the hook, so the hook goes through it.
- Lower the loop until it sits all the way down on the hook and the coat takes its own weight.
- Open the gripper and draw it back away from the coat.

Check: the second coat is hanging right and hanging clear, beside the first coat and not on top of it. The first coat has not been knocked off its hook. If the two coats hang on one hook, or one

has come off, the **left gripper** closes on the loop again, lifts it just off, and puts it on the right hook.

Expected state: both coats hang on the two coat hooks, side by side and clear of each other. The coat spot is empty. The bag hook is empty. Both grippers are back and clear.

Step 3: Put the first shoe on the rack

Goal: one shoe stands upright on the rack shelf, and the shoe spot holds one shoe.

3.1 Lift the first shoe

- The **left gripper** closes on the shoe nearer the left end at the shoe spot, around the middle of the shoe.
- Lift it straight up off the bench, just off, and bring it forward into the front lane. Keep it upright, sole down.

Check: the shoe hangs upright and level in the **left gripper**, sole down. The other shoe has not moved. If the shoe hangs upside down or by a lace, the **left gripper** puts it back down on the bench, closes on the middle again, and lifts it again.

3.2 Carry it to the rack

- The **left gripper** carries the shoe along the front lane to the left end of the bench in a low carry, keeping it upright the whole way.
- Do not carry it across the back of the bench and do not carry it over the bowl.
- Then lower it down to the rack shelf, clear of the bench and clear of the mat.

Check: the shoe is over the rack shelf, still upright, sole down. No part of the arm rests on the rack or on the bench. If the rack moves, stop and report it.

3.3 Stand it on the shelf

- The **left gripper** lowers the shoe straight down onto the rack shelf, at the back half of the shelf, leaving room in front of it for the second shoe.
- Stand it flat on its sole, fully on the shelf, with its toe facing out to the front.
- Open the gripper and draw it back away from the shoe.

Check: the shoe is standing right on the rack. It stands flat on its sole, fully on the shelf, toe to the front, and it does not fall or rock when the gripper lets go. If it tips or hangs over the front edge, the **left gripper** closes on it again, lifts it just off the shelf, and stands it down again.

Expected state: one shoe stands on the rack shelf. One shoe lies at the shoe spot. Both coats still hang. Both grippers are back and clear.

Step 4: Pair the second shoe beside it

Goal: both shoes stand paired on the rack shelf, and the shoe spot is empty.

4.1 Lift the second shoe

- The **left gripper** closes on the shoe left at the shoe spot, around the middle of the shoe.
- Lift it straight up off the bench, just off, and bring it forward into the front lane. Keep it upright, sole down.

Check: the shoe hangs upright and level in the **left gripper** and the shoe spot is empty. If it hangs by a lace, the **left gripper** puts it back down, closes on the middle, and lifts it again.

4.2 Carry it to the rack

- The **left gripper** carries the shoe along the front lane to the left end of the bench in a low carry, keeping it upright the whole way.
- Do not carry it across the back of the bench and do not carry it over the bowl.
- Then lower it down to the rack shelf, in front of the shoe that is already there.

Check: the shoe is over the free part of the shelf, still upright, and it is not over the first shoe. If it hangs over the first shoe, the **left gripper** moves it forward before lowering it.

4.3 Stand it beside the first

- The **left gripper** lowers the shoe straight down onto the rack shelf, next to the first shoe.
- Stand it flat on its sole, facing the same way as the first shoe, with the two heels level and a finger's width between them.
- Open the gripper and draw it back away from the shoe.

Check: the two shoes are paired. They stand side by side, both toes to the front, heels level, with a small gap between them, and neither one rocks when the gripper lets go. The first shoe has not been pushed off the shelf. If they are crossed, one behind the other, or touching hard, the **left gripper** closes on the second shoe again, lifts it just off the shelf, and stands it down again.

Expected state: both shoes stand paired on the rack shelf. The shoe spot is empty. The coat spot is empty. Both grippers are back and clear.

Step 5: Put the keys in the bowl

Goal: the whole key set lies in the bowl, and the key spot is empty.

5.1 Lift the keys

- The **right gripper** closes on the key ring at the key spot.
- Lift the whole set straight up off the bench, just off, and bring it forward into the front lane. Do not swing it. Do not drag it along the bench.

Check: the whole set hangs from the ring in the **right gripper** and the key spot is empty. No key is left on the bench. If a key slid off the ring, stop and report it.

5.2 Carry them to the bowl

- The **right gripper** carries the keys along the front lane to the right end of the bench in a low carry, keeping them low and still.
- Then move them straight back to the bowl. Do not swing them over the bowl.
- Hold them over the middle of the bowl, just above its rim.

Check: the keys hang still over the middle of the bowl. They are not over the bench top and not over the bag. If they swing, the **right gripper** holds them still and lets them settle before lowering them.

5.3 Lower them in

- The **right gripper** lowers the keys straight down into the bowl until they rest on its floor.
- Open the gripper and draw it back away from the bowl.
- Do not touch the rim of the bowl and do not push the bowl.

Check: the keys are in the bowl. The whole set lies inside, below the rim, with no key hanging over the edge and none on the bench. The bowl has not moved. If a key hangs over the rim, the **right gripper** closes on the ring again, lifts the set just off, and lowers it in again.

Expected state: the keys lie in the bowl. The key spot is empty. Both coats hang and both shoes are paired. Both grippers are back and clear.

Step 6: Hang the bag

Goal: the bag hangs on the bag hook, and the bench top is clear.

6.1 Lift the bag by its strap

- The **right gripper** closes on the strap of the bag at the bag spot, at the middle of the strap.
- Lift the bag straight up until its bottom just clears the bench, and bring it forward into the front lane. Keep it hanging straight from the strap.

Check: the bag hangs straight down from its strap in the **right gripper** and the bag spot is empty. The strap is not caught under the bag. If the bag hangs at an angle or by its body, the **right gripper** puts it back down flat, lays the strap clear, closes on the middle of the strap, and lifts it again.

6.2 Carry it to the hook

- The **right gripper** carries the bag along the front lane to the bag hook in a low carry, keeping it hanging straight the whole way.
- Do not carry it across the back of the bench and do not carry it over the bowl.
- Bring it in front of the hook, then lift the strap straight up, level with the hook.

Check: the bag hangs straight and still in front of the bag hook, clear of the coats and clear of the bowl. If it swings, the **right gripper** holds it still and lets it settle before going on.

6.3 Put the strap on the hook

- The **right gripper** moves the strap straight back over the hook, so the hook goes through it.
- Lower the strap until it sits all the way down on the hook and the bag takes its own weight.
- Open the gripper and draw it back away from the bag.

Check: the bag is hanging right and hanging clear. The strap sits all the way over the hook, the bag hangs straight down, it does not swing or lift off, and it does not rest on the bench top or on the bowl. If the strap sits on the tip of the hook or the bag rests on the bench, the **right gripper** closes on the strap again, lifts it just off the hook, and puts it on again.

Expected state: two coats and the bag hang on the three hooks. The keys lie in the bowl. Both shoes stand paired on the rack. The bench top is clear. Both grippers are back and clear.

Step 7: Square the mat

Goal: the mat lies flat on the floor in front of the bench, with its near edge straight and parallel to the bench front edge.

The mat is only ever pushed. It is not lifted, folded, or straightened any other way in this step.

7.1 Look at the mat

- Look at the mat from the front, with the front edge of the bench in the same view.
- See which end of the near edge is further from the bench, and by how much. That end's corner is the one to push, and the gripper on that corner's side pushes it: the **left gripper** for the left corner, the **right gripper** for the right corner.

Check: the whole mat is in view, it lies flat, and no corner is folded under. If a corner is folded under, stop and report it. The mat is not lifted to fix it.

7.2 Push it square

- The gripper on that corner's side closes and comes down to the mat's edge, close to the floor, at the corner picked.
- Push the mat flat along the floor in short pushes, until its near edge runs straight across, parallel to the front edge of the bench.
- If the mat still runs off square, the other gripper pushes the other corner the same way.

Check: the near edge of the mat is square to the bench, and both ends of that edge are the same distance from the bench. The mat still lies flat, with no bunching and no corner folded under. If it has bunched up, the same gripper pushes it flat again from the same edge.

7.3 Set the gap

- Look at the strip of floor between the bench front edge and the mat's near edge.
- The **right gripper** pushes the mat forward or back at the middle of its near edge, in short pushes, until that strip is the same width all the way across.

- Draw the gripper back away from the mat and up off the floor.

Check: the mat is square, it lies flat, and the gap to the bench is even end to end. No part of an arm rests on the mat or on the floor. If the mat turned off square while the gap was set, the gripper on that corner's side pushes the corner again until it runs straight.

Expected state: the mat lies flat and square in front of the bench. Nothing else has moved. Both grippers are back and clear.

Step 8: End the episode

Goal: the entryway is reset, and both arms are safely home.

- Confirm the coats: one hangs on each coat hook, straight down, side by side, clear of each other and clear of the bench.
- Confirm the shoes: both stand on the rack shelf, upright, facing the same way, heels level, paired, and fully on the shelf.
- Confirm the keys: the whole set lies in the bowl, below the rim, with no key outside it.
- Confirm the bag: it hangs on the bag hook, straight down, clear of the coats, the bowl, and the bench top.
- Confirm the bench top is clear, with nothing left at the coat spot, the shoe spot, the key spot, the bag spot, or the front lane, and the mat lies flat and square with an even gap to the bench.
- Confirm the fixtures stand where they started, nothing is on the floor but the shoes on the rack and the mat, and the base has not moved: it is still square to the bench and still locked.
- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded. The base stays parked and locked through the reset, and is only pushed away by hand once the bench is set for the next episode.

1. Take both coats off their hooks. Lay them flat at the coat spot the next episode's config puts them, front left for Config L, front middle for Config M, the right end past the bag for Config R, one on top of the other, each with its loop up and clear.
2. Take the bag off its hook. Lay it at the bag spot, past the keys, with its strap laid out clear and not under the bag.
3. Take the keys out of the bowl. Lay the whole set at the key spot.
4. Take both shoes off the rack. Lay them loose at the shoe spot, front middle for Config L and R, front left for Config M, apart from each other and not paired.
5. Push the mat off square again, so it starts crooked, and check it lies fully flat with no corner folded under.
6. Wipe the bench top and dry it. Wipe the bowl inside and the rack shelf, and dry them.

7. Check all three hooks are empty, tight on the board, and not bent.
8. Check the bowl is empty and stands at the back right of the bench top.
9. Check the shoe rack is empty, standing square on the floor at the left end of the bench, and clear of the bench.
10. Replace a coat whose loop is torn, worn loose, or will not stand up when the coat lies flat.
11. Replace the bag if its strap is torn or will not stay over the hook in a correct hang.
12. Replace a shoe that will not stand upright on its own sole on a flat shelf.
13. Replace the mat if it will not slide flat, if it bunches when it is pushed, or if a corner will not lie down.
14. Clear anything that ended up on the floor and dry the floor before the next episode.
15. Check the base is still square, centered, and locked, and push it firmly once by hand to confirm it does not roll. If it has moved, park it again by the Base positioning steps.
16. Run the Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the coat spot: the violations below were written for Config L (coat stack at the front left, lifted and hung by the left gripper). The arm-role cues in them will be rewritten later to cover all three configs; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the grippers do does not match the config on the bench. A gripper reaches across the bench for a coat; a coat is handed over in Config L or M, or is not handed over in Config R; the right gripper hangs a coat; the coat stack or the shoes are not where that config puts them; or the wrong IF line is followed in Step 1 or 2.
- **SOP rule broken:** Steps 1.1 and 2.1, the same-side rule and the hand-over rule. The gripper on the coat stack's side lifts the coat, the left gripper hangs it, the hand-over

happens only in Config R, and the IF line followed is the one for the config on the bench.

- **Coaching note:** look where the coat stack lies before anything is touched, then follow that config's IF lines through Steps 1 and 2.

Violation: Base moved during the episode

- **Visible cue:** the bench shifts in the frame, the bench edge changes angle in the frame, or the base rolls, creeps, or turns at any time after recording starts.
- **SOP rule broken:** Step 8 and the Base positioning rule it confirms, the base is parked and locked before recording and does not move at all during the episode.
- **Coaching note:** park it, lock it, push-test it. If the park is wrong, fix it before recording, never during.

Violation: Lifted high or hit the hook board

- **Visible cue:** a coat, a shoe, the keys, or the bag is lifted well above the bench instead of just off what it lies on, or a wrist or forearm knocks, scrapes, or rests on the hook board. The only lifts that may go up are the loop and the strap going onto their hooks.
- **SOP rule broken:** Steps 1.1, 1.2, 2.1, 2.2, 3.1, 3.2, 4.1, 4.2, 5.1, 5.2, 6.1, and 6.2, lift just off, come forward into the front lane, and carry low and level.
- **Coaching note:** up a little, forward, then along. Only the loop and the strap go up, and only at the hook.

Violation: Carried outside the front lane

- **Visible cue:** a carry goes across the back of the bench or over the bowl instead of along the front lane, or a thing is put down in the lane in the middle of a carry.
- **SOP rule broken:** Steps 1.2, 2.2, 3.2, 4.2, 5.2, and 6.2, every carry goes along the front lane, and nothing is ever put down in it.
- **Coaching note:** the lane is the road. Nothing crosses the back of the bench, and nothing parks in the lane.

Violation: Leaned on or moved a fixture

- **Visible cue:** a gripper, wrist, or forearm rests on the bench, the hook board, a hook, the bowl, or the shoe rack, or the bowl slides, the rack tips or slides, or the hook board moves.
- **SOP rule broken:** Steps 1 to 7, nothing leans on, pushes, drags, or lifts the bench, the hook board, a hook, the bowl, or the shoe rack.
- **Coaching note:** the arm holds itself up. If something moves when it is touched, it was touched too hard.

Violation: Held by the wrong part

- **Visible cue:** a gripper closes on a coat's sleeve, hem, collar, or body instead of its loop, on the bag's body instead of its strap, on a shoe's lace or tongue instead of the middle of the

shoe, or on a single key instead of the ring.

- **SOP rule broken:** Steps 1.1, 2.1, 3.1, 4.1, 5.1, and 6.1, coats by the loop, the bag by the strap, a shoe by its middle, the keys by the ring.
- **Coaching note:** one hold for each thing, every time. The right hold is the one that hangs straight.

Violation: Dragged instead of lifted

- **Visible cue:** a coat, a shoe, the keys, or the bag slides along the bench top instead of coming straight up off it before it moves.
- **SOP rule broken:** Steps 1.1, 2.1, 3.1, 4.1, 5.1, and 6.1, every pick lifts the thing straight up off the surface first.
- **Coaching note:** up first, then along. A dragged thing pulls the next one with it.

Violation: Two things moved at once

- **Visible cue:** both coats come off the stack together, both shoes are held at the same time, one thing is still held while another is picked up, or a thing is passed from one gripper to the other outside the Config R coat hand-over.
- **SOP rule broken:** Steps 1 to 7, only one thing is held at a time, it is let go before the next one starts, and nothing is passed between grippers except the coat in Config R.
- **Coaching note:** one thing, all the way to its place, then let go. Then the next one.

Violation: Not hanging right

- **Visible cue:** a loop or the strap sits on the tip of a hook, hangs half over it, or comes off when the gripper lets go, or a coat or the bag is left on the bench instead of on its hook.
- **SOP rule broken:** Steps 1.3, 2.3, and 6.3, the loop or the strap sits all the way over the hook and takes its own weight before the gripper lets go.
- **Coaching note:** all the way over, then watch it take its own weight before opening the gripper.

Violation: Not hanging clear

- **Visible cue:** a hung coat or the bag rests on the bench top, on the bowl, or on the wall, a sleeve or a strap lies across the bench, or the two coats hang tangled or both on one hook.
- **SOP rule broken:** Steps 1.3, 2.3, and 6.3, each thing hangs free of the bench, the wall, and the thing next to it.
- **Coaching note:** three hooks, three things, nothing touching. If it leans on something, it is not hung.

Violation: Shoes not paired

- **Visible cue:** at the end of Step 4 the two shoes stand crossed, one behind the other, one on top of the other, facing different ways, with heels well out of line, or one shoe is left off the rack.
- **SOP rule broken:** Steps 3.3 and 4.3, both shoes stand side by side, facing the same way, heels level, with a finger's width between them.
- **Coaching note:** side by side, same way, heels in line. A pair is read in one look.

Violation: Shoe not standing right

- **Visible cue:** a shoe lies on its side, stands on the front edge of the shelf with part hanging over, rocks when the left gripper lets go, or falls off the rack.
- **SOP rule broken:** Steps 3.3 and 4.3, stand it flat on its sole, fully on the shelf, and watch it settle before letting go.
- **Coaching note:** sole down, fully on, then let go. A shoe that rocks falls after the arm leaves.

Violation: Keys not in the bowl

- **Visible cue:** the keys are dropped from above the bowl, land on the bench, hang over the rim, bounce out, or a key ends up outside the bowl.
- **SOP rule broken:** Step 5.3, lower the whole set straight down into the bowl until it rests on the floor of the bowl.
- **Coaching note:** lower them in, do not drop them in. Keys bounce.

Violation: Mat lifted or bunched

- **Visible cue:** a gripper lifts the mat off the floor, digs under it, folds it, or pushes it so hard that it bunches up or rolls.
- **SOP rule broken:** Steps 7.2 and 7.3, the mat is pushed flat along the floor and is never lifted.
- **Coaching note:** short pushes at the edge, close to the floor. The mat stays down.

Violation: Mat left crooked

- **Visible cue:** at the end of Step 7 the near edge of the mat does not run parallel to the bench front edge, or the gap to the bench is clearly wider at one end than the other.
- **SOP rule broken:** Steps 7.2 and 7.3, the near edge runs straight across, and the gap is even end to end.
- **Coaching note:** look along the edge, not at the middle. If the gap changes, the mat is not square.

Violation: Done in the wrong order

- **Visible cue:** a shoe moves before both coats hang, the keys move before both shoes are paired, the bag moves before the keys are in the bowl, or the mat is pushed before the

bench top is clear.

- **SOP rule broken:** Steps 1 to 7, coats, then shoes, then keys, then bag, then the mat.
- **Coaching note:** one thing to its place before the next one starts. The mat is always last.

Violation: Wrong arm used

- **Visible cue:** the right gripper carries a shoe or stands it on the rack, the left gripper lifts the keys or the bag, or the left gripper sets the mat gap.
- **SOP rule broken:** Steps 3 to 7, the left gripper owns the shoes, the right gripper owns the keys, the bag, and the mat gap, and each corner of the mat is pushed by the gripper on its side.
- **Coaching note:** left arm to the rack, right arm to the bowl and the bag hook.

Violation: Bench left with something on it

- **Visible cue:** at the end of the episode a coat, a shoe, the keys, or the bag is still on the bench top, or something is left standing in the front lane.
- **SOP rule broken:** Step 8, the bench top is clear at the end, with nothing left at any spot and nothing in the lane.
- **Coaching note:** the bench is empty when the work is done. If it is on the bench, it is not put away.

Violation: Something dropped or knocked over

- **Visible cue:** a coat, a shoe, the keys, or the bag slips out of a gripper, falls on the bench or on the floor, or is knocked over by an arm going past, or the bowl is tipped.
- **SOP rule broken:** Steps 1 to 7, one thing moves at a time, and each one is put down or let go under control.
- **Coaching note:** close on the hold all the way before lifting, and put it down before letting go.

Violation: Failed check not retried

- **Visible cue:** a check in a step clearly fails and the episode carries on with no retry: a loop sits on the tip of a hook, a shoe rocks on the shelf, a key hangs over the bowl rim, or the mat still runs crooked.
- **SOP rule broken:** Steps 1 to 7, do the retry written under each failed check before moving on.
- **Coaching note:** a check is only worth doing if the retry is done. Fix it where it happened.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a coat, a shoe, the keys, or the bag out of place, the bench top not clear, the mat crooked, or an arm away from home.

- **SOP rule broken:** Step 8 (confirm the coats, the shoes, the keys, the bag, the bench, the mat, the fixtures, and the locked base; return both arms home; then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot see a hook, inside the bowl, the rack shelf, or the near edge of the mat**, so nobody can tell if a thing was placed right.
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Base fault:** a brake that releases on its own, or a caster that seizes so the base cannot be parked square.
- **Defective object:** a coat with a torn or missing loop, a loop that will not stand up when the coat lies flat, or a coat too heavy for its hook; a bag with a torn strap, a strap too short or too wide for the hook, or a bag that spills when it hangs; a shoe that will not stand upright on its own sole, or one wet or dirty underneath; a key ring that comes open, or a key loose off the ring at the start; a mat that will not slide flat, that bunches or rolls when pushed at its edge, or with a corner that will not lie down. Replace it before the next episode.
- **Fixture fault:** a hook board that moves, turns, sags, or comes loose under a correctly placed loop or strap; a hook bent, too short, or too smooth to hold a loop or a strap; a shoe rack that tips, slides, or sags under a correctly placed shoe, or a shelf too narrow for the two shoes side by side; a bowl that slides under a correct lowering, that is too small or too shallow for the whole key set, or that is cracked; a floor that is wet, sticky, or uneven, so the mat will not slide or will not lie flat.
- **Reach fault:** a place turns out to be too far for its arm with the base parked correctly, so a coat, a shoe, the keys, the bag, the bowl, a hook, the rack shelf, or the mat cannot be reached without extending the arm to a limit.

Annotation subtasks (from SOP)

1. Lift the first coat off the bench by its loop
2. Hand the coat over to the left gripper (Config R only)
3. Carry the first coat along the front lane to coat hook one
4. Put the first coat's loop on the hook
5. Lift the second coat off the bench by its loop
6. Carry the second coat along the front lane to coat hook two
7. Put the second coat's loop on the hook

8. Lift the first shoe off the bench
9. Carry the first shoe along the front lane to the rack
10. Stand the first shoe on the rack shelf
11. Lift the second shoe off the bench
12. Carry the second shoe along the front lane to the rack
13. Stand the second shoe beside the first and pair them
14. Lift the keys off the bench by the ring
15. Lower the keys into the bowl
16. Lift the bag off the bench by its strap
17. Carry the bag along the front lane to the bag hook
18. Put the bag's strap on the hook
19. Push the mat square to the bench
20. Set the gap between the mat and the bench
21. Return both arms home and end the episode